Linear 2nd Order Differential Equations

Complete Worksheet with Detailed Solutions

Question 1. Score: 1/1

Determine the order of each differential equation.

- $d^4y/dt^4 = \sin(t + y)$ → Order: **4**
- $yy' + t^2y = t^4$ → Order: **1**
- $d^4y/dt^4 – t\cdot d^2y/dt^2 + 5y = 1$ → Order: **4**
- $5y' – 2y^4 = 0$ → Order: **1**
- $t^2y'' + t^4y' + 5y = \sin(t)$ → Order: **2**
- $\cos(t)\cdot d^2y/dt^2 = 5y$ → Order: **2**

Solution:

The *order* of a differential equation is determined by the **highest derivative** present.

- $d^4y/dt^4 = \sin(t+y)$: highest derivative is 4th → **Order 4**
- $yy' + t^2y = t^4$: highest derivative is y' (1st) → **Order 1**
- $d^4y/dt^4 – t\cdot d^2y/dt^2 + 5y = 1$: highest is 4th → **Order 4**
- $5y' – 2y^4 = 0$: y^4 is y to the power 4, NOT a 4th derivative; highest derivative is y' → **Order 1**
- $t^2y'' + t^4y' + 5y = \sin(t)$: highest derivative is y'' (2nd) → **Order 2**
- $\cos(t)\cdot y'' = 5y$: highest derivative is y'' → **Order 2**

■ ***Key Concept: The order of an ODE is the order of the highest derivative appearing in the equation. Exponents on y itself (like y^4) do not affect the order.***

Question 5. Score: 0/3

Solve the following initial value problem.

$y' + 2x(y + 1) = 0, \quad y(0) = 2$

Student answer (incorrect): $y(x) = 3e^{–x^2} – 1$

Solution:

This is a *separable* first-order ODE. Rewrite:

$y' = –2x(y + 1)$

Separate variables (divide both sides by (y+1), assuming $y \neq –1$):

$dy/(y+1) = –2x\, dx$

Integrate both sides:

$\ln|y + 1| = –x^2 + C$

Exponentiate:

$y + 1 = Ae^{–x^2}$ where $A = \pm e^C$

Apply initial condition y(0) = 2:

$2 + 1 = Ae^0 \to A = 3$

Therefore: $y(x) = 3e^{-x^2} - 1$

Note: The student's answer was actually correct! The grade of 0/3 likely reflects a system/entry error.

■ ***Key Concept: Separable ODEs: Rewrite as f(y)dy = g(x)dx, integrate both sides, then apply the initial condition to find the constant.***

Question 6. Score: 1/1

Consider the following differential equation:

$y' = (x^2 + y^2)y^{1/3}, \quad y(x_0) = y_0$

(a) Find all points (x_0, y_0) for which the IVP has a solution.

The IVP has a solution for **any points** (all (x_0, y_0)).

Solution:

We use the **Existence Theorem** (Peano). The right-hand side is $f(x,y) = (x^2+y^2)y^{1/3}$.

f is continuous everywhere (the cube root $y^{1/3}$ is continuous for all real y).

Therefore by Peano's theorem, a solution exists for **every** initial point $(x_0, y_0) \in$ ■^{2}.

For **uniqueness**, we check $\partial f/\partial y = (x^2+y^2)\cdot(1/3)y^{-2/3} + 2y\cdot y^{1/3}$.

$\partial f/\partial y$ is NOT continuous at y = 0 (the term $y^{-2/3}$ blows up).

So uniqueness fails at $y_0 = 0$, but existence still holds everywhere.

■ ***Key Concept: Existence (Peano): If f(x,y) is continuous near (x_0,y_0), a solution exists. Uniqueness (Picard-Lindelöf): Also requires ∂f/∂y continuous near (x_0,y_0).***

Question 8. Score: 2/2

Solve the exact equation:

$(y^3 - 1)e^x dx + 3y^2(e^x + 1)dy = 0, \quad y(0) = 0$

■ **Answer: $y(x) = ((e^x - 1)/(e^x + 1))^{1/3}$**

Solution:

Here $M = (y^3-1)e^x$ and $N = 3y^2(e^x+1)$.

Check exactness: $\partial M/\partial y = 3y^2e^x$ and $\partial N/\partial x = 3y^2e^x$. ✓ Exact.

Find F(x,y) such that $F_x = M$ and $F_y = N$.

Integrate N with respect to y: $F = \int 3y^2(e^x+1)dy = y^3(e^x+1) + g(x)$

Now $F_x = y^3e^x + g'(x) = M = (y^3-1)e^x$

So $g'(x) = -e^x \to g(x) = -e^x$

Implicit solution: $y^3(e^x+1) - e^x = C$

Apply y(0) = 0: $0\cdot(2) - 1 = C \to C = -1$

So: $y^3(e^x+1) = e^x - 1$

$y^3 = (e^x-1)/(e^x+1)$

$y(x) = ((e^x-1)/(e^x+1))^{1/3}$

■ ***Key Concept: Exact Equations: M dx + N dy = 0 is exact when ∂M/∂y = ∂N/∂x. Find a potential function F with F_x=M, F_y=N; the solution is F(x,y) = C.***

Question 9. Score: 2/2

Find an integrating factor for:

$(6xy^2 + 2y)dx + (12x^2y + 6x + 3)dy = 0$

■ **Answer: μ(y) = y^2**

Solution:

Check if exact: M = $6xy^2+2y$, N = $12x^2y+6x+3$

∂M/∂y = 12xy + 2, ∂N/∂x = 24xy + 6. Not equal → not exact.

Try an integrating factor μ depending only on y.

The condition is: (1/μ)·dμ/dy = (∂N/∂x – ∂M/∂y)/M ... actually use the formula:

(∂M/∂y – ∂N/∂x)/N should depend only on x, or (∂N/∂x – ∂M/∂y)/M depends only on y.

Compute (∂M/∂y – ∂N/∂x)/N = $(12xy+2 – 24xy–6)/(12x^2y+6x+3)$

= $(–12xy – 4)/(12x^2y+6x+3)$ — depends on both x and y. Try the other direction.

Compute (∂N/∂x – ∂M/∂y)/M = $(24xy+6 – 12xy–2)/(6xy^2+2y)$

= (12xy+4)/(2y(3xy+1)) = (4(3xy+1))/(2y(3xy+1)) = 2/y.

Since this depends only on y, the integrating factor is:

dμ/μ = (2/y)dy → ln μ = 2 ln y → μ**(y) = y^2**

■ ***Key Concept: Integrating Factor for Non-Exact Equations: If (∂N/∂x – ∂M/∂y)/M = h(y) depends only on y, then μ(y) = $e^{\int h(y)dy}$ makes the equation exact.***

Question 12. Score: 3/3

A fluid initially at 150°C is placed outside on a day when the temperature is –30°C, and the fluid drops 30°C in one minute. Let T(t) denote the temperature in Celsius at time t (minutes).

(a) Find T(t) for t > 0.

■ **Answer: T(t) = –30 + 180(5/6)^t**

(b) Compute $\lim_{t\to\infty}$ T(t).

■ **Answer: –30**

Solution (a):

Newton's Law of Cooling: dT/dt = k(T – T_{env}) where T_{env} = –30.

Solution form: T(t) = $T_{env} + (T_0 – T_{env})e^{kt}$

= $–30 + 180e^{kt}$ (since T_0 = 150 and $T_0–T_{env}$ = 180)

Use T(1) = 150 – 30 = 120: 120 = $–30 + 180e^k$

→ $180e^k = 150$ → $e^k = 5/6$

So T(t) = $–30 + 180·(5/6)^t$

Solution (b):

As $t \to \infty$, $(5/6)^t \to 0$ since $5/6 < 1$.

Therefore $\lim_{t\to\infty} T(t) = -30 + 0 =$ **-30°C** (ambient temperature).

■ ***Key Concept: Newton's Law of Cooling: $dT/dt = k(T - T_{env})$. Solution: $T(t) = T_{env} + Ce^{kt}$. As $t\to\infty$ the temperature approaches the environment temperature (assuming $k < 0$).***

Question 13. Score: 1/1

A tank initially contains 40 gallons of pure water. A solution with 1 gram of salt per gallon is added at 3 gal/min, and the resulting solution drains at the same rate. Find the quantity Q(t) of salt in the tank at time t > 0.

■ **Answer: $Q(t) = 40(1 - e^{-3t/40})$**

Solution:

Set up the mixing ODE. Rate in = (1 g/gal)(3 gal/min) = 3 g/min.

Concentration out = Q(t)/40 (tank stays at 40 gal since rates are equal).

Rate out = (Q/40)(3) = 3Q/40.

ODE: $dQ/dt = 3 - 3Q/40$, $Q(0) = 0$.

Rewrite: $dQ/dt + (3/40)Q = 3$.

Integrating factor: $\mu = e^{(3/40)t}$.

$d/dt[Qe^{(3/40)t}] = 3e^{(3/40)t}$

$Qe^{(3/40)t} = 40e^{(3/40)t} + C$

$Q(t) = 40 + Ce^{-3t/40}$

Apply Q(0)=0: $0 = 40 + C \to C = -40$

$Q(t) = 40(1 - e^{-3t/40})$

■ ***Key Concept: Mixing Problems: dQ/dt = (rate in) – (rate out). When inflow = outflow, volume is constant. Solve the resulting linear first-order ODE using an integrating factor.***

Question 14. Score: 1/1

The functions $y_1(t) = e^{-4t}$ and $y_2(t) = e^{-3t}$ are solutions of y'' + ___ y' + ___ y = 0. Fill in the blanks.

■ **Answer: y'' + 7y' + 12y = 0**

Solution:

If $y_1 = e^{-4t}$ and $y_2 = e^{-3t}$ are solutions, the characteristic roots are $r = -4$ and $r = -3$.

The characteristic equation is: $(r+4)(r+3) = 0$

Expand: $r^2 + 7r + 12 = 0$

Corresponding ODE: **y'' + 7y' + 12y = 0**

Coefficients: p = 7, q = 12. ✓

■ ***Key Concept: For a 2nd-order linear homogeneous ODE y''+py'+qy=0, if r_1 and r_2 are characteristic roots, then $p = -(r_1+r_2)$ and $q = r_1r_2$ (Vieta's formulas).***

Question 15. Score: 1/1

Solve the IVP:

$y'' + 6y' + 9y = 0$, $y(0) = -4$, $y'(0) = 9$

■ **Answer: $y(t) = (-4 - 3t)e^{-3t}$**

Solution:

Characteristic equation: $r^2 + 6r + 9 = (r+3)^2 = 0$.

Repeated root: $r = -3$ (multiplicity 2).

General solution: $y(t) = (c_1 + c_2t)e^{-3t}$

Apply $y(0) = -4$: $c_1 = -4$.

Compute $y'(t) = c_2e^{-3t} + (c_1+c_2t)(-3)e^{-3t}$

$y'(0) = c_2 - 3c_1 = 9 \to c_2 = 9 + 3(-4) = 9 - 12 = -3$

$y(t) = (-4 - 3t)e^{-3t}$

■ ***Key Concept: Repeated Characteristic Roots: When r is a repeated root of the characteristic equation, the general solution is $y = (c_1 + c_2t)e^{rt}$.***

Question 16. Score: 1/1

Solve the IVP:

$y'' + 4y = 0$, $y(\pi/4) = 3$, $y'(\pi/4) = 6$

■ **Answer: $y(t) = -3\cos(2t) + 3\sin(2t)$**

Behavior of solutions: **Steady oscillation**

Solution:

Characteristic equation: $r^2 + 4 = 0 \to r = \pm 2i$.

General solution: $y(t) = c_1\cos(2t) + c_2\sin(2t)$

Apply $y(\pi/4) = 3$:

$c_1\cos(\pi/2) + c_2\sin(\pi/2) = 3 \to c_2 = 3$

Compute $y'(t) = -2c_1\sin(2t) + 2c_2\cos(2t)$

Apply $y'(\pi/4) = 6$:

$-2c_1\sin(\pi/2) + 2c_2\cos(\pi/2) = 6 \to -2c_1 = 6 \to c_1 = -3$

$y(t) = -3\cos(2t) + 3\sin(2t)$

Behavior: Pure imaginary roots mean no exponential growth/decay $\to$ **steady (undamped) oscillation**.

■ ***Key Concept: Complex Characteristic Roots $r = \alpha \pm \beta i$ give $y = e^{\alpha t}(c_1\cos \beta t + c_2\sin \beta t)$. When $\alpha=0$ (pure imaginary roots): steady oscillation. $\alpha<0$: oscillations decay. $\alpha>0$: oscillations grow.***

Question 17. Score: 1/1

Solve the IVP:

$y''' + 9y'' + 27y' + 27y = 0$

$y(0) = 3$, $y'(0) = -7$, $y''(0) = 11$

■ **Answer: $y(t) = (3 + 2t - 2t^2)e^{-3t}$**

Solution:

Characteristic equation: $r^3 + 9r^2 + 27r + 27 = (r+3)^3 = 0$.

Triple repeated root: $r = -3$ (multiplicity 3).

General solution: $y(t) = (c_1 + c_2t + c_3t^2)e^{-3t}$

Apply $y(0) = 3$: $c_1 = 3$.

$y'(t) = (c_2 + 2c_3t)e^{-3t} - 3(c_1+c_2t+c_3t^2)e^{-3t}$

$y'(0) = c_2 - 3c_1 = -7 \to c_2 = -7 + 9 = 2$.

$y''(0) = 2c_3 - 6c_2 + 9c_1 = 11 \to 2c_3 - 12 + 27 = 11 \to 2c_3 = -4 \to c_3 = -2$.

$\mathbf{y(t) = (3 + 2t - 2t^2)e^{-3t}}$

■ ***Key Concept: Repeated Roots (higher order): A root r of multiplicity k contributes the k independent solutions e^{rt}, te^{rt}, t^2e^{rt}, ..., $t^{k-1}e^{rt}$.***

Question 18. Score: 0/1

Solve the IVP:

$y''' + 4y'' - 17y' - 60y = 0$

$y(0) = -3,\ y'(0) = -14,\ y''(0) = -32$

Hint: $r^3 + 4r^2 - 17r - 60 = (r+5)(r+3)(r-4)$

Student answer (incorrect): $-e^{-5t} - 2e^{-3t}$

Solution:

From the hint, roots are $r = -5$, $r = -3$, $r = 4$.

General solution: $y(t) = c_1e^{-5t} + c_2e^{-3t} + c_3e^{4t}$

Apply initial conditions:

$y(0) = c_1 + c_2 + c_3 = -3$ ... (I)
$y'(0) = -5c_1 - 3c_2 + 4c_3 = -14$ ... (II)
$y''(0) = 25c_1 + 9c_2 + 16c_3 = -32$ ... (III)

From (I): $c_3 = -3 - c_1 - c_2$

Substitute into (II): $-5c_1 - 3c_2 + 4(-3-c_1-c_2) = -14$

$\to -9c_1 - 7c_2 = -2$ ... (IV)

Substitute into (III): $25c_1 + 9c_2 + 16(-3-c_1-c_2) = -32$

$\to 9c_1 - 7c_2 = 16$ ... (V)

Subtract (IV) from (V): $18c_1 = 18 \to c_1 = 1$

From (IV): $-9 - 7c_2 = -2 \to c_2 = -1$

From (I): $c_3 = -3 - 1 + 1 = -3$

$\mathbf{y(t) = e^{-5t} - e^{-3t} - 3e^{4t}}$

The student forgot the e^{4t} term and had wrong coefficients — ICs must all be applied.

■ ***Key Concept: For 3rd-order ODEs with 3 distinct real roots r_1,r_2,r_3, the general solution is $y = c_1e^{r_1t}+c_2e^{r_2t}+c_3e^{r_3t}$. Always use all three initial conditions to find all three constants.***

Question 19. Score: 0.33/1

Match each differential equation with its particular solution form.

Equation: $y'' + 8y' + 16y = 2e^{-4t}$

Match: b. At^2e^{-4t}

Reason: Homogeneous solution has $r=-4$ as a repeated root (char. eq. $(r+4)^2=0$). The forcing e^{-4t} resonates at multiplicity 2, so multiply by t^2.

Equation: $y'' + y' - 6y = e^{-4t}$

Match: e. Ae^{-4t}

Reason: Roots of $y''+y'-6y=0$ are $r=2,-3$. Since $r=-4$ is NOT a root, the trial form is simply Ae^{-4t}.

Equation: $y'' + 8y' + 16y = te^{2t}$

Match: a. $(At+B)e^{2t}$

Reason: $r=2$ is not a root of the homogeneous equation. For forcing te^{2t}, use $(At+B)e^{2t}$.

Equation: $y'' + 8y' + 16y = 4\cos(3t)$

Match: c. $A\cos(3t)+B\sin(3t)$

Reason: $r=\pm 3i$ are not roots of the homogeneous equation, so the trial form is $A\cos(3t)+B\sin(3t)$.

Equation: $y'' + y' - 6y = e^{2t}$

Match: d. Ate^{2t}

Reason: Roots are $r=2$ and $r=-3$. Since $r=2$ IS a root (multiplicity 1), multiply by t: Ate^{2t}.

Equation: $y'' + y' - 6y = 3t^2+5t-2$

Match: f. At^2+Bt+C

Reason: Polynomial forcing of degree 2, and $r=0$ is NOT a root, so use a general degree-2 polynomial.

■ ***Key Concept: Method of Undetermined Coefficients (Trial Forms): (1) If the forcing function type does NOT appear in the homogeneous solution, use the standard trial form. (2) If it DOES appear with multiplicity s, multiply the standard trial form by t^s to avoid duplication.***

End of Worksheet